

Exercise 4B

Problem 1 The number of kilometers traveled, X , until a fall occurs in a certain cycling race can be modeled with an exponential distribution with probability density function $f(x) = \lambda \exp^{-\lambda x}$. With $\lambda = 0.01$ the average number of falls per kilometer.

- i) Simulate 1000 realisations of X , i.e. 1000 distances between consecutive falls, using uniform random numbers to generate observations from the exponential distribution.
- ii) Graphically compare the simulated numbers with the theoretical distribution.

Problem 2 The angular size in degrees, X , of cosmic textures leaving their imprint on the Cosmic Microwave Background would be a random variable X with probability density function $f(x) = Kx^{-3}$, where $K > 0$ is a real constant and $x > 1$ for physical reasons.

- i) Calculate the value of K .
- ii) Generate 1000 random numbers following the distribution of X .
- iii) Graphically compare the simulated numbers with the theoretical distribution.

Problem 3 We throw darts at a target and model the distance to the target as a random vector $Z = (Z_1, Z_2)$ where $Z_i \sim \mathcal{N}(0, 1)$, $i = 1, 2$.

- i) Simulate 1000 vectors Z using the function `rnorm` and plot them.
- ii) What distribution will the polar coordinates $Z = (r, \theta)$ follow? Plot the histograms of r and θ .
- iii) Simulate the polar coordinates of Z using uniform random numbers.
- iv) Transform the polar coordinates into Z_1 and Z_2 to obtain normal random numbers.

Problem 4 Use the `photutils` library to perform the following tasks:

- i) Simulate an image of size 500×500 with 15 point sources with random positions, random intensity between 600 and 1000. The parameters σ_x, σ_y that determine the size of the sources will be random between 4 and 5 pixels.
- ii) Generate Gaussian noise with standard deviation $\sigma_N = 1000$ and add it to the image. Visualize the noisy image and observe whether you can still distinguish the sources.
- iii) Convolve the noisy image with a Gaussian kernel of $\sigma = 4$ pixels.
- iv) Search for the sources in the convolved image with the `find_peaks` function and detection threshold `threshold = 5 \sigma` where σ is the standard deviation of the convolved image.
- v) Display the convolved image, marking the original sources and the found sources, and comment on the results obtained.
- vi) What happens if you change the detection threshold to 3σ ?